Soln: Suppose A has p vectors and B has q vectors and wLog let $p<q$. $A=[a_1, a_2, \ldots, a_p]$ and $B=[b_1, \ldots, b_q]$

By defn. of bases, both A and B contain linearly independent vectors and each of them span W. $A, B \in W$.

By Prob 27, $\because q>p$, $\therefore$ The columns of B are linearly dependent which is a contradiction. Similarly, $p>q$ is False.

$\therefore p=q$ □